

Multi-Agent Contract Design with a Budget*

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Abstract

We study a multi-agent contract design problem with moral hazard. In our model, each agent exerts costly effort towards an individual task at which it may either succeed or fail, and the principal, who wishes to encourage effort, has an exclusive-use budget that it can use to reward the agents. A motivating application is crowdsourcing for innovation, where a fixed budget is provided to a crowdsourcing platform to use for rewarding participants based on their submissions. Our main contribution is to introduce a novel class of contracts, which we call Luce contracts, and show that there is always a Luce contract that is optimal. A Luce contract assigns weights and a priority order to the agents, and distributes the entire budget among the highest-priority successful agents in proportion to their weights. Furthermore, we characterize effort profiles that can be implemented by Luce contracts and show that Luce contracts offer a way to mitigate the uncertainty in total payments compared to alternative contracts—such as piece-rate or bonus-pool contracts—suggesting their desirability even in environments without budget constraints.

1 Introduction

We study a contract design problem for a principal interacting with many independent agents, as is the case for example in crowdsourcing environments. While the principal could,

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in theory, treat each agent independently and offer an individually optimal piece-rate contract, this incentive scheme has several undesirable features, including significant variability in total payments and the risk of excessively large payouts. In practice, contracts in such environments are often not piece-rate but instead structured as competitions, where the principal sets aside a fixed budget and commits to distributing the budget among a subset of well-performing agents. For example, crowdsourcing platforms like Wazoku and HeroX invite organizations to pose their unsolved problems as Challenges to a crowd of solvers, outlining criterion for a submission to be successful and specifying a budget for awarding successful submissions. In this paper, we model the contract design problem in the presence of budgetary constraints, and further, formalize the sense in which the competition structure of these contracts helps mitigate the uncertainty in total payments.

In our model, each agent exerts costly effort towards a task, in which it may independently succeed or fail, and the principal, who is budget-constrained, can commit to a contract which defines the payment to each agent as a function of the set of agents who are successful. For convenience, we assume that the principal’s budget is exclusive-use, meaning that any unused portion of the budget provides no additional value to the principal. The assumption is appropriate when the budget is provided as a restricted-use fund or when agents are rewarded with probability shares for an indivisible prize. It is also reasonable in environments where the budget is small in comparison to the benefit associated with successful submissions, as might be the case in crowdsourcing for innovation. The principal’s problem is then to find a contract, which describes how the budget is distributed among the agents for each outcome, so as to incentivize agents to exert higher effort towards being successful.

We first show that, even though there might be multiple optimal contracts, any such contract must be *successful-get-everything* (SGE): it rewards only successful agents and distributes the entire budget among them. In particular, contracts such as piece-rate or bonus-pool contracts, which under certain outcomes distribute only a fraction of the budget to successful agents, can never be optimal in this setting. Intuitively, the principal should be able to improve upon contracts that leave money on the table by using that leftover money to increase the rewards of some successful agent, thus incentivizing them to exert greater effort.¹ We further show that any SGE contract is optimal for some monotone objective of the principal. In other words, the set of maximal implementable equilibria — i.e., the coordinate-wise maximal elements of the equilibrium set — is exactly the set of equilibria of SGE contracts.

Our main contribution is then to introduce a subclass of SGE contracts, called Luce contracts, and to show that there is always a Luce contract that is optimal. A Luce contract is defined by assigning a weight to each agent and a (weak) priority order over the set of agents,

¹This intuition is slightly misleading, since equilibrium forces imply that there isn’t necessarily an obvious way to distribute this left-over budget and induce higher effort from all agents. Nevertheless, we show that there is always some way to do this, and this implies that any optimal contract must be SGE.

so that the entire budget is distributed among the highest-priority successful agents in proportion to their weights. We show that Luce contracts are minimally sufficient to implement the maximal set, and thus, the principal can always optimize over Luce contracts. This result provides for a significant reduction in the complexity of the design problem, narrowing the search space from a $\Theta(n2^n)$ -dimensional space of complex and difficult-to-interpret contracts to an $(n - 1)$ -dimensional space of simple and easily interpretable contracts. We apply our results to the special case of two agents with quadratic costs, where our analysis uncovers an interesting comparative static: under the optimal contract, the principal strengthens incentives for the agent it cares about more as the environment becomes more difficult, through an increase in either agent’s cost or a decrease in the budget.

Lastly, we discuss the desirability of Luce contracts for implementation in more standard contract design environments, where the principal incurs a direct cost from rewarding the agents but is not budget-constrained. In such environments, the principal has access to a wide range of contracts, including piece-rate and bonus-pool contracts, for the purpose of inducing any desired effort profile. We characterize the set of effort profiles that can be implemented via Luce contracts with some budget and show that, for any such profile, the total payment under any implementing contract is a mean-preserving spread of the total payment under the corresponding Luce contract. It follows that Luce contracts minimize both the maximum total payment the principal might need to make and the variability in total payment for these effort profiles. With these desirable properties, we believe Luce contracts offer a useful alternative to commonly studied contracts, like piece-rate and bonus-pool contracts, for inducing effort in general multi-agent contract design environments.

Related literature

There is a vast literature on principal-agent problems under moral hazard. In the canonical model with a single agent (Holmström, 1979; Grossman and Hart, 1992; Mirrlees, 1976), the principal offers a wage contract that defines the agent’s payment as a function of its observed output, and then the agent chooses some unobserved action (effort) that determines the distribution over outputs. A key finding is that optimal contracts reward the agent for output realizations that are informative about the target level of effort (informativeness principle), and may therefore be non-monotone in output. There has since been significant work studying variants of this single-agent model incorporating flexible actions (Georgiadis, Ravid, and Szentes, 2024), multiple tasks (Holmstrom and Milgrom, 1991; Bond and Gomes, 2009), bounded payments (Jewitt, Kadan, and Swinkels, 2008), combinatorial actions (Dütting, Ezra, Feldman, and Kesselheim, 2022), and informationally robust design (Carroll, 2015; Zhang, 2023). See Georgiadis (2022), Holmström (2017), and Lazear (2018) for recent surveys of this literature.

Our paper contributes to the literature studying a principal contracting with multiple agents. In these settings, agents may be rewarded not only based on their absolute performance but also on their relative performance or even the joint performance of the group.

When agents’ observed outputs are subject to a common shock or involve subjective evaluations, the optimal contract may involve pay for relative performance (Green and Stokey, 1983; Lazear and Rosen, 1981; Malcomson, 1986, 1984; Mookherjee, 1984; Nalebuff and Stiglitz, 1983; Imhof and Kräkel, 2014). And joint performance evaluation may be optimal in the presence of negatively correlated outputs (Fleckinger, 2012), complementarities in production (Alchian and Demsetz, 1972), incentives to help (Itoh, 1991), and unknown production environment (Kambhampati, 2024). Other related work on multi-agent contract design explores moral hazard in teams (Holmstrom, 1982; Winter, 2004; Battaglini, 2006; Babaioff, Feldman, Nisan, and Winter, 2012; Halac, Lipnowski, and Rappoport, 2021; Dai and Toikka, 2022), computational challenges in finding optimal contracts (Castiglioni, Marchesi, and Gatti, 2023; Dütting, Ezra, Feldman, and Kesselheim, 2023; Ezra, Feldman, and Schlesinger, 2024), and design problems tailored to more specific environments (Halac, Kartik, and Liu, 2017; Haggiag, Oren, and Segev, 2022; Baiman and Rajan, 1995). In contrast to this literature, our work focuses on a principal contracting with multiple independent agents, where standard single-agent contracts would be optimal but are rendered infeasible due to budgetary constraints on the principal.²

2 Model

There is a principal and n risk-neutral agents. Each agent $i \in [n]$ is assigned an independent task, in which it may succeed or fail. The agents choose how much effort to exert towards succeeding in their respective tasks, captured by their choice of success probabilities: if agent i chooses success probability $p_i \in [0, 1]$, it incurs a cost $c_i(p_i)$ in doing so, where $c_i : [0, 1] \rightarrow \mathbb{R}$ is differentiable, strictly increasing, and strictly convex, with $c'_i(0) = 0$. The cost functions are common knowledge.

The principal would like to incentivize the agents to maximize the probability that they succeed in their tasks. Formally, we assume that the principal’s preference over success probability profiles is represented by a continuous, strictly increasing objective function $V(p_1, \dots, p_n)$. For example, if the principal obtains a profit of v_i if agent i succeeds in its task, and it cares about maximizing expected profit, its preference may be represented by the objective function

$$V(p) = \sum_{i=1}^n v_i p_i.$$

Alternatively, if the principal gets a constant positive payoff as long as there is at least one successful agent, and 0 otherwise (as may be the case in a crowdsourcing for innovation

²There is also a related literature in contests, where the contract specifies a distribution of the budget based on how agents rank in terms of their output (Fang, Noe, and Strack, 2020). However, in our model, the principal observes only whether agents succeed or fail, not their exact effort levels or rankings. Thus, the reward schemes typically analyzed in contest settings are not applicable to our framework.

example), its preference may be represented by

$$V(p) = 1 - \prod_{i=1}^n (1 - p_i).$$

In order to incentivize the agents, the principal can design a contract mapping observed outcomes to a reward for each agent. We assume that the principal has an exclusive-use budget $B > 0$ that can be used for the purpose of rewarding the agents, and the principal gets no value from unused portions of the budget. Formally, the principal commits to a contract which specifies the fraction of the budget each agent receives depending on the observed outcome.

Definition 1. A *contract* is a function $f = (f_1, \dots, f_n) : 2^{[n]} \rightarrow \mathbb{R}_+^{[n]}$ such that

- $f_i(S) \geq 0$ (limited liability) and
- $\sum_{j \in [n]} f_j(S) \leq 1$ (budget constraint)

for each $i \in [n]$ and $S \subseteq [n]$.

Under the contract f , if S is the set of agents who succeed, then each agent i receives the reward $f_i(S) \cdot B$. For instance, a natural contract in this environment would be to split the budget equally amongst agents who are successful:

$$f_i(S) = \begin{cases} \frac{1}{|S|}, & i \in S \neq \emptyset \\ 0, & \text{otherwise.} \end{cases}$$

Note that, as in this contract, each agent's reward may depend not only on whether she succeeds, but also on the success of the other agents. We will denote by $\mathcal{F}$ the set of contracts. Throughout the paper, we will work with several subclasses of contracts, and we will in general denote by $\mathcal{F}_\phi$ the set of contracts with property ϕ .

A contract $f \in \mathcal{F}$ defines a normal-form game between the n agents, in which each agent chooses $p_i \in [0, 1]$ and agent i 's payoff under the profile $p = (p_1, \dots, p_n)$ is its expected reward minus the cost of choosing p_i . Formally,

$$u_i(p) = \mathbb{E}_p[f_i(S) \cdot B] - c_i(p_i) = \sum_{S \subseteq [n]} \Pr_p^{[n]}(S) \cdot f_i(S) \cdot B - c_i(p_i)$$

where

$$\Pr_p^{[n]}(S) = \prod_{i \in S} p_i \prod_{j \in [n] \setminus S} (1 - p_j).$$

Notice that the expected reward is linear in p_i , and since the cost is convex, the utility is strictly concave in p_i . It thus follows that pure-strategy Nash equilibrium exists. We will denote by $E : \mathcal{F} \rightrightarrows [0, 1]^n$ the equilibrium correspondence, so $E(f)$ is the set of pure-strategy

equilibria for the contract f and $E^{-1}(p)$ is the set of contracts for which p is an equilibrium. We further denote by $\mathcal{E}$ the set of implementable profiles:

$$\mathcal{E} := \{p \in [0, 1]^n : p \in E(f) \text{ for some } f \in \mathcal{F}\}.$$

The principal's problem is to choose a contract f that induces an equilibrium $p \in E(f)$ that maximizes its objective V . As is standard, we assume that when a contract has multiple equilibria, the principal can select its most preferred equilibrium.³ Hence, the principal's problem is to maximize its objective V over the set of implementable profiles $\mathcal{E}$. As we show, $\mathcal{E}$ is compact, and since V is strictly increasing, any maximizer of V must be a maximal element of $\mathcal{E}$ with respect to the standard coordinatewise ordering. We denote by $\mathcal{P}$ the set of maximal equilibrium profiles in $\mathcal{E}$. Formally,

$$\mathcal{P} = \{p \in \mathcal{E} : \forall q \in \mathcal{E}, q_i \geq p_i \forall i \in [n] \Rightarrow q = p\}.$$

The principal's problem thus reduces to maximizing its objective V over the set of maximal profiles $\mathcal{P}$:

$$\max_{p \in \mathcal{P}} V(p).$$

Small budget

Notice that if the budget is sufficiently large ($B \geq \sum_{i \in [n]} c'_i(1)$), the principal can induce any effort profile $q \in [0, 1]^n$ using the following piece-rate contract:

$$f_i(S) = \begin{cases} \frac{1}{B} \cdot c'_i(q_i), & i \in S \\ 0, & \text{otherwise.} \end{cases}$$

Indeed, agent i 's expected utility in this case is $p_i \cdot c'_i(q_i) - c_i(p_i)$ which is uniquely maximized at $p_i = q_i$. It follows that $\mathcal{E} = [0, 1]^n$, $\mathcal{P} = \{(1, 1, \dots, 1)\}$, and the problem is trivial.

Hence, we focus on the more interesting case where the principal's budget is small. In particular, we assume that the principal's budget is small enough that it could not induce even one of the agents to succeed with probability 1.

Assumption 1. The principal's budget $B < \min\{c'_1(1), c'_2(1), \dots, c'_n(1)\}$.

Assumption 1 ensures that in equilibrium, no agent chooses $p_i = 1$. We note that our results extend to situations with a larger budget as long as we restrict attention to equilibria that are interior.

³For an illustration of multiplicity, suppose $B \geq \sum_{i \in [n]} c'_i(1)$ and consider the bonus-pool contract under which agent i 's reward is $c'_i(1)$ if all agents succeed and 0 otherwise. For this contract, it is easy to verify that both $p = (1, 1, \dots, 1)$ and $p = (0, 0, \dots, 0)$ are Nash equilibria. That said, if the budget is sufficiently small and the cost functions are sufficiently convex, we can use the diagonally strict concavity condition of Rosen (1965) to show that every contract has a unique equilibrium.

3 Main results

In this section, we identify and characterize a family of contracts whose equilibria coincide precisely with the set of maximal equilibria.

Consider any contract $f \in \mathcal{F}$. At any effort profile $p \in [0, 1]^n$, agent i 's utility can be conveniently expressed as

$$u_i(p) = (\mathbb{E}_p[f_i(S) \mid i \in S] \cdot p_i + \mathbb{E}_p[f_i(S) \mid i \notin S] \cdot (1 - p_i)) \cdot B - c_i(p_i). \quad (1)$$

Since u_i is strictly concave in p_i , agent i has a unique best response characterized by the first-order condition:

$$c'_i(p_i) = \max\{0, (\mathbb{E}_p[f_i(S) \mid i \in S] - \mathbb{E}_p[f_i(S) \mid i \notin S]) \cdot B\}. \quad (2)$$

We denote this unique best response by $b_i(f, p_{-i})$.

Equation (2) makes clear that rewarding agents when they fail reduces their incentive to exert effort, and suggests that the principal should avoid using contracts that reward unsuccessful agents. We introduce this class of contracts formally:

Definition 2. A contract $f \in \mathcal{F}$ is *failures-get-nothing* (FGN) if $f_i(S) = 0$ whenever $i \notin S$.

It turns out that using a contract which is not FGN is wasteful in a very particular sense. Given any such contract and an effort profile p it implements, it is possible to scale the payments down at every outcome to form an FGN contract which also implements p . As a consequence, FGN contracts are not only sufficient, but offer the cheapest alternative for inducing any implementable profile.

Proposition 1. *For any implementable profile p , there exists an FGN contract g which implements it. Moreover, the expected total payment at p under g is minimal:*

$$\mathbb{E}_p\left[\sum_{i \in [n]} g_i(S)\right] \cdot B = \sum_{i \in [n]} p_i c'_i(p_i).$$

Going forward, without loss of generality, we restrict attention to FGN contracts. Under any FGN contract f and effort profile p , Equation (2) simplifies to:

$$c'_i(p_i) = \mathbb{E}_p[f_i(S) \mid i \in S] \cdot B. \quad (3)$$

This equation shows clearly that an agent's incentive to exert effort increases in their expected share of the budget if they succeed. Since the principal's budget is exclusive-use, meaning that it does not derive any value from leftover budget, this suggests that the principal should always distribute the entire budget among successful agents.

Definition 3. A contract f is *successful-get-everything* (SGE) if

- $f_i(S) = 0$ whenever $i \notin S$ and
- $\sum_{i \in S} f_i(S) = 1$ whenever $S \neq \emptyset$.

Given any implementable profile p , if the FGN contract f which induces it is not SGE, there must be some leftover budget under some outcomes. Observe that the most the principal can give in expected rewards at p is $\mathbb{P}_p[S \neq \emptyset] \cdot B$, and the actual amount it gives in expected rewards is $\mathbb{E}_p[\sum_i f_i(S)] \cdot B$. We denote by $z(p)$ this gap between the upper bound and the actual amount given in rewards, and note that it depends only on p :

$$\begin{aligned} z(p) &= \mathbb{P}_p[S \neq \emptyset] \cdot B - \mathbb{E}_p \left[\sum_i f_i(S) \right] \cdot B \\ &= \left[1 - \left(\prod_{i \in [n]} (1 - p_i) \right) \right] \cdot B - \sum_{i \in [n]} p_i c'_i(p_i), \end{aligned}$$

where the second term follows from Proposition 1. For any $p \in \mathcal{E}$, $z(p)$ provides a useful robust measure of the leftover budget from inducing p using any FGN contract $f \in E^{-1}(p)$.

Corollary 1. *If p is implementable, then $z(p) \geq 0$. Moreover, if p is implementable by an SGE contract, then $z(p) = 0$.*

Now suppose effort profile p is an equilibrium of a SGE contract. If q is an effort profile that coordinate-wise dominates p , we can show that $z(q) < z(p) = 0$, and so by Corollary 1, q is not implementable. It thus follows that any equilibrium of any SGE contract is maximal.

Interestingly, maximal equilibria can only be induced by SGE contracts. Intuitively, if p is an equilibrium of a FGN contract which is not SGE, the leftover budget could potentially be used to incentivize higher effort. While one might suspect that increasing agent i 's share $f_i(S)$ for some $i \in S$ (if $\sum_{i \in S} f_i(S) < 1$) might do the trick, the equilibrium forces imply that such a transformation does not necessarily lead to a dominant outcome.⁴ In general, even though there is not an obvious way to transform these contracts so that they induce dominant outcomes, we show there always exists some way.

Theorem 1. *Suppose the contract f implements the effort profile p . Then, p is maximal if and only if f is an SGE contract.*

We now outline our proof to show that if f is not SGE, then $p \in E(f)$ cannot be maximal. Consider first a contract f which never uses the entire budget—i.e. $(\sum_i f_i(S) < 1$ for all $S \subset$

⁴For an illustration, consider an example with $n = 2$, $c_i(p_i) = \frac{1}{2}C_i p_i^2$, and $B = 1$ (so Assumption 1 requires $C_1, C_2 > 1$). Suppose $f \in \mathcal{F}$ is such that it awards $\beta_i \in [0, 1]$ to agent i if and only if i is the only successful agent. Using Equation (3), one can verify that f has a unique equilibrium $p = (p_1, p_2)$ with $p_i = \frac{\beta_i(C_{-i} - \beta_{-i})}{C_1 C_2 - \beta_1 \beta_2}$. From here, it is easy to check that increasing β_i would increase p_i , but it may also be accompanied by a decrease in p_{-i} .

$[n]$). Fix a small $\epsilon > 0$. For each agent i , consider scaling agent i 's reward at every outcome by a factor $t_i(\epsilon)$ so that the first-order condition (Equation (3)) is satisfied at $p'_i = p_i + \epsilon$ given that every other agent j chooses $p'_j = p_j + \epsilon$. As $\epsilon \rightarrow 0$, each $t_i(\epsilon) \rightarrow 1$, so by continuity, we can choose ϵ small enough so that the contract defined by scaling the payments by $(t_i(\epsilon))$ remains feasible, and implements the profile p' , which dominates p . This is the first important observation: while the equilibrium effect of increasing payments arbitrarily may not be to increase all agents' efforts, it is possible to control the equilibrium effects by appropriately scaling the payments.

The next important observation is that if f is a contract which fails to use the entire budget often enough—in particular, if f is a contract such that for every agent i , there is some outcome S where i succeeds (i.e. $i \in S$) and $\sum_j f_j(S) < 1$ —then there is also a contract g which implements p and which never uses the entire budget. This leaves only the case that f always uses the entire budget when some agents succeed, but not when other agents succeed. Denoting this latter set by κ_p , we show that through appropriate transfers of rewards for agents across outcomes, and taking convex combinations of such contracts, one can construct a new contract h which implements p such that (i) under h , agents in κ_p are rewarded only at outcomes where no agent outside κ_p succeeds, and (ii) h never uses the entire budget when only agents in κ_p succeed. Since the effort choices of agents in κ_p do not affect the incentives of agents outside it, we can then apply the scaling argument above to the agents in κ_p alone to obtain a profile that dominates p .

By Theorem 1, since contracts like piece-rate (where each agent i gets a fixed reward if she is successful and 0 otherwise) or bonus-pool (where each agent gets a nonzero share of the budget only if all agents succeed) are not SGE, they never induce maximal equilibria. Thus, the result suggests that in environments where the principal operates with an exclusive-use budget, exhausting the budget using successful-get-everything contracts creates stronger incentives than promoting teamwork through bonus-pool contracts or independent performance assessment through piece-rate contracts.

While these simpler, well-known contracts should be avoided in favor of SGE contracts, the set of all SGE contracts itself lacks clear structure, being $\Theta(n2^n)$ -dimensional. Consequently, optimizing over SGE contracts may still pose a computationally difficult problem, and the resulting optimal contract may be complex to interpret and implement in practice. Now, the set of maximal equilibria lies on the boundary of the equilibrium set and is thus at most $(n - 1)$ -dimensional. This dimensionality gap suggests that for any given maximal equilibrium, there are potentially many SGE contracts that implement it, which raises the question: is there a simpler, low-dimensional, structured subclass of SGE contracts that the principal can use? Weighted contracts, which are both structured and dimensionally aligned, provide a natural and promising candidate.

Definition 4. A contract f is a *weighted* (W) contract if there exist weights $(\lambda_1, \dots, \lambda_n)$

with $\lambda_i > 0$ such that

$$f_i(S) = \begin{cases} \frac{\lambda_i}{\sum_{j \in S} \lambda_j}, & \text{if } i \in S \\ 0, & \text{otherwise} \end{cases}$$

A weighted contract assigns weights to the agents, and distributes the entire budget among successful agents in proportion to their weights. As the following result shows, these simple contracts do in fact essentially span the whole set of maximal equilibria.

Theorem 2. *For every maximal equilibrium p , there is a unique contract in the closure of weighted contracts which implements p .*

The closure of weighted contracts also has a relatively simple structure. We refer to these contracts, containing weighted contracts and their limit points, as Luce contracts. In addition to assigning weights to the agents, a general Luce contract also specifies a priority ordering over the agents. The ordering arises because some agents' weights may grow without bound relative to others' along a sequence of weighted contracts.⁵

Definition 5. A contract f is a *Luce contract* if there exist weights $(\lambda_1, \dots, \lambda_n)$ with $\lambda_i > 0$ and a weak ordering $\succsim$ on the agents such that

$$f_i(S) = \begin{cases} \frac{\lambda_i}{\sum_{j \in \text{Top}_{\succsim}(S)} \lambda_j}, & \text{if } i \in \text{Top}_{\succsim}(S) \\ 0, & \text{otherwise} \end{cases}$$

where $\text{Top}_{\succsim}(S) = \{i \in S : i \succsim j \ \forall j \in S\}$.

Luce contracts reward each highest-priority successful agent with a fraction of the budget proportional to her weight. Weighted contracts are exactly the Luce contracts in which all agents share the same priority. The ordering gives the principal a way to prioritize among the agents: a successful agent receives nothing whenever a higher-priority agent also succeeds.

The choice of terminology is motivated by the analogy to the selection rules that satisfy Luce's choice axiom (Luce, 1959). Just as Luce rules have the property that the relative odds of selecting one item over another from a pool is unaffected by which other items are in the pool, Luce contracts have the property that the reward of a successful agent relative to another successful agent is independent of which other agents succeed.

Lemma 1. *A contract f is in the closure of weighted contracts if and only if it is a Luce contract.*

⁵Consider the sequence of weighted contracts with weights $(1, 1, m)$. As $m \rightarrow \infty$, agent 3's weight grows without bound relative to those of agents 1 and 2. In the limit contract, agent 3 receives the entire budget whenever it succeeds; when it fails, the budget is split evenly among whichever of agents 1 and 2 succeed.

We now outline our proof of Theorem 2. The existence part follows immediately from Proposition 3 in Section 4 below, which provides a complete characterization of effort profiles that can be implemented by Luce contracts. In particular, Proposition 3 implies that any equilibrium which can be implemented using an SGE contract can be implemented using a Luce contract, and by Theorem 1, every maximal equilibrium can be implemented by an SGE contract. For uniqueness, suppose f and g are distinct weighted contracts which both implement p , and let j be an agent whose ratio of weights $\frac{\lambda_j^f}{\lambda_j^g}$ is minimal. Then $f_j(S) \leq g_j(S)$ for all $S \subseteq [n]$ with strict inequality for some S , so the best response of agent j to p_{-j} must be strictly lower under f than under g , and hence p cannot be an equilibrium for both f and g . A similar but more technical argument extends to Luce contracts and is deferred to the appendix.

It follows from Theorem 2 that the principal can always just optimize its objective over the class of Luce contracts.

Corollary 2. *For every maximal equilibrium p , there is a unique Luce contract which implements p . Thus, for any strictly increasing continuous objective $V(p)$,*

$$\sup_{f \in \mathcal{F}} \sup_{p \in E(f)} V(p) = \max_{f \in \mathcal{F}_{\text{Luce}}} \max_{p \in E(f)} V(p).$$

Thus, Theorem 2 provides a significant reduction in the complexity of the principal's problem, reducing the search space from a $(2^{n-1}(n-2) + 1)$ -dimensional space of difficult to interpret contracts to an $(n-1)$ -dimensional space of easily interpretable contracts.

Despite this reduction in complexity, it remains difficult to explicitly solve for the optimal Luce contract for a given objective in general. We are able to solve the problem for the special case where there are two agents with quadratic costs.

Proposition 2. *Suppose $n = 2$, $c_i(p_i) = \frac{1}{2}C_i p_i^2$ with $C_i > B$, and $V(p_1, p_2) = v_1 p_1 + v_2 p_2$ with $v = \frac{v_1}{v_2} \geq 1$. Then, the optimal contract, defined by λ_1 , takes the form*

$$f_i(S) = \begin{cases} 0, & \text{if } i \notin S \\ 1, & \text{if } S = \{i\} \\ \lambda_i, & \text{if } S = \{1, 2\} \end{cases},$$

where

$$\lambda_1 = \begin{cases} \frac{1}{2}, & \text{if } v = 1 \\ \frac{-(vC_2 + C_1 - vB) + \sqrt{(v^2C_2 + C_1 - vB)(C_2 + C_1 - B)}}{B(v-1)}, & \text{if } v \in (1, \frac{C_1C_2 + BC_1 - B^2}{C_1C_2 - BC_2}) \\ 1, & \text{if } v \geq \frac{C_1C_2 + BC_1 - B^2}{C_1C_2 - BC_2} \end{cases}$$

and $\lambda_2 = 1 - \lambda_1$. It follows that λ_1 is weakly increasing in v , C_1 , and C_2 , and weakly decreasing in B .

Proposition 2 identifies the principal’s relative valuation ($v = \frac{v_1}{v_2}$) of the two agents as a key determinant of the optimal contract’s structure. When $v_1 = v_2$, the equi-split contract ($\lambda_1 = \lambda_2$) is optimal, regardless of the agents’ cost parameters C_1 and C_2 . In general, an agent’s share is increasing in the relative value that the principal places on that agent’s success, and if this value is large enough, the entire budget is awarded to the more valued agent. This latter observation illustrates the importance of the priority ordering to the principal: when v is large (or small), the optimal contract is a Luce contract which is not a weighted contract.

Interestingly, the share awarded to the higher-valued agent is also increasing in the difficulty of the environment. Specifically, when $v_1 > v_2$, the share λ_1 is increasing in C_1 , increasing in C_2 , and decreasing in B : as either agent becomes less productive, or the budget becomes smaller, the optimal contract assigns a greater share to the more-valued agent.

It is natural to ask how far these insights extend. We can show that the robustness of the optimal contract to cost heterogeneity under equal valuations does not generalize beyond the two-agent case.⁶ We are unable to say more on whether the optimal contract generally strengthens incentives for a more highly valued agent as the environment becomes more difficult; we view this as a nuanced and novel insight that presents an interesting question for future work.

We conclude this section with two remarks highlighting how our results on the optimality of Luce contracts may extend to other settings where the principal is budget-constrained, but the budget is not exclusive-use or the outcomes are not binary.

Remark 1 (Relaxing exclusive-use). *The exclusive-use assumption—that the principal derives no value from unspent budget—is convenient but not essential for the optimality of Luce contracts. Suppose instead that the principal obtains a residual value $\alpha \geq 0$ per unit of unspent budget, so that its effective payoff at profile p is $V(p) - \alpha C(p)$, where $C(p)$ denotes the expected total payment at p . By Proposition 1, the minimum expected payment across all contracts implementing any profile p is $\sum_i p_i c'_i(p_i)$, and this minimum is achieved by any FGN contract—in particular, by Luce contracts. This reduces the principal’s problem to choosing a contract with budget B that maximizes*

$$V(p) - \alpha \sum_{i \in [n]} p_i c'_i(p_i).$$

Whenever this objective is strictly increasing in p on the set of implementable profiles, the optimum is attained at a maximal profile. By Theorem 2, there is a unique Luce contract which implements it. To illustrate, for $V(p) = \sum_i v_i p_i$ and $c_i(p_i) = \frac{1}{2} C_i p_i^2$, the condition

$$\min_i \frac{v_i}{2C_i} > \alpha$$

⁶When $n = 3$, $B = 1$, and $c_i(p_i) = \frac{1}{2} C_i p_i^2$ with $C_1 = 3.32, C_2 = C_3 = 1.81$, the contract maximizing $p_1 + p_2 + p_3$ induces the effort profile $p = (0.2, 0.4, 0.4)$ which is not an equilibrium of the equi-split contract.

ensures that the objective is increasing in p , and thus, there must be a Luce contract that is optimal. The optimality of Luce contracts therefore extends beyond the exclusive-use case ($\alpha = 0$) to settings in which the principal places a positive but sufficiently small value on unspent budget, so that the effective objective remains increasing in p .

Remark 2 (Beyond binary outcomes). *Although our model assumes binary outcomes for each agent, it can also be interpreted within a more general framework by restricting attention to a simple and natural class of target-based contracts. Formally, suppose each agent $i \in [n]$ chooses an unobserved effort level $e_i \in \mathbb{R}_+$, which induces a probability distribution over a set of observable outcomes X_i . Consider contracts in which the principal specifies a target $t_i \in X_i$ for each agent, so that the agent is deemed to have succeeded if the realized outcome is at least t_i and failed otherwise. Under standard assumptions—such as X_i being ordered and higher effort inducing distributions that first-order stochastically dominate those under lower effort—our results can be interpreted as characterizing optimal target-based contracts within this richer environment. Specifically, for any fixed profile of targets $(t_1, \dots, t_n)$, there are again multiple ways in which the principal can distribute the budget depending upon who meets the target. From our results, an optimal way to do so is via a Luce contract.*

4 Luce contracts without budget constraints

In the previous section, we established that Luce contracts are optimal when the principal is budget-constrained. However, this assumption departs from standard contract-design models, where the principal is not budget-constrained but incurs a direct cost from rewarding the agents that it weighs against the benefits of inducing higher effort. In this section, we argue that Luce contracts offer a desirable alternative for inducing effort even in the absence of a budget constraint.

In the absence of a budget constraint, the principal has access to a broad range of contracts for implementing any desired effort profile $q \in (0, 1)^n$. For instance, the principal could offer each agent $i \in [n]$ an independent piece-rate contract that pays $c'_i(q_i)$ upon success and nothing otherwise. Alternatively, the principal could offer a bonus-pool contract which pays $\frac{q_i c'_i(q_i)}{\prod_{i \in [n]} q_i}$ to agent $i \in [n]$ if all agents succeed and nothing otherwise. Perhaps, the principal could also set aside a budget $B = \frac{\sum_{i \in [n]} q_i \cdot c'_i(q_i)}{\mathbb{P}_q[S \neq \emptyset]}$ and design a Luce contract that implements q . Unfortunately, this may not always be feasible, as the structure of Luce contracts introduces constraints on effort profiles that they can implement.

To see why, fix any agent $i \in [n]$. Their expected payment under a contract that induces q must equal $q_i c'_i(q_i)$. For a Luce contract with budget B , the agent's expected payment is maximized when it receives the entire budget B whenever it succeeds, yielding an upper bound of $\mathbb{P}_q(i \in S) \cdot B$. Consequently, a necessary condition for q to be implementable by a Luce contract is that for each $i \in [n]$, $q_i c'_i(q_i) \leq \mathbb{P}_q(i \in S) \cdot B$. The same logic applies to

any subset I : the expected total payment to agents in I under a Luce contract is bounded above by $\mathbb{P}_q(S \cap I \neq \emptyset) \cdot B$, giving the necessary condition $\sum_{i \in I} q_i c'_i(q_i) \leq \mathbb{P}_q(S \cap I \neq \emptyset) \cdot B$. The following result shows that these inequalities are also sufficient, thereby characterizing effort profiles which can be implemented via a Luce contract.

Proposition 3. *Suppose $p \in (0, 1)^n$. There exists a Luce contract $f \in \mathcal{F}_{\text{Luce}}$ with budget $B > 0$ that implements p if and only if for all $I \subset [n]$,*

$$\sum_{i \in I} p_i \cdot c'_i(p_i) \leq \mathbb{P}_p[S \cap I \neq \emptyset] \cdot B, \quad (4)$$

where $B = \frac{\sum_{i \in [n]} p_i \cdot c'_i(p_i)}{\mathbb{P}_p[S \neq \emptyset]}.$

We now outline our proof that if p satisfies (4), then it can be implemented by a Luce contract. As we show, for any Luce contract f with budget B , there is an agent i that has an incentive to deviate upward from p_i ($b_i(f, p_{-i}) > p_i$) if and only if there is an agent j that has an incentive to deviate downward ($b_j(f, p_{-j}) < p_j$). Consequently, there must be at least one agent i for which $b_i(f, p_{-i}) \geq p_i$, and so $Z(f) = \max_{i \in [n]} (c'_i(b_i(f, p_{-i})) - c'_i(p_i))$ must be nonnegative. Since p is an equilibrium for f if and only if $Z(f) = 0$, it suffices to show that the infimum of $Z(f)$ over the (compact) space of Luce contracts is 0. To illustrate the argument that the minimum is indeed 0, suppose that the minimizer f is a weighted contract with $Z(f) > 0$. Then there is some agent j who has an incentive to deviate downwards—i.e. for whom $c'_j(b_j(f, p_{-j})) - c'_j(p_j) < 0$. Increasing λ_j^f by a small amount keeps this term negative while strictly reducing the best response for all other agents, thereby strictly reducing $Z(f)$. Hence, if the minimizer f is a weighted contract, p must be an equilibrium of f . A more involved version of this argument shows that this also holds if the minimizer is any Luce contract.

Figure 1 illustrates the set of implementable effort profiles for two identical agents. As the figure suggests, every symmetric profile is implementable by a Luce contract. This holds more generally: with n identical agents, for every symmetric effort profile, there is a Luce contract that implements it. On the other hand, Luce contracts cannot in general accommodate very asymmetric effort profiles—for any fixed budget, there are upper and lower bounds on the agents' effort levels that can be induced.

Proposition 3 provides a valuable tool for determining whether a desired effort profile can be implemented by a Luce contract. In a standard contract design environment, the principal wishes to induce an effort profile maximizing $V(p) - C(p)$, where $C(p) = \sum_{i \in [n]} p_i c'_i(p_i)$ is the expected total payment under any contract implementing p . The principal can solve this problem and verify whether the solution satisfies (4); if so, a Luce contract implements it. We illustrate this in the following example.

Example 1. *Suppose $n = 2$, $c_i(p_i) = \frac{1}{2} C_i p_i^2$ and $V(p) = v_1 p_1 + v_2 p_2$ (as in Proposition 2). The principal's desired effort profile maximizes $v_1 p_1 + v_2 p_2 - C_1 p_1^2 - C_2 p_2^2$. Within the*

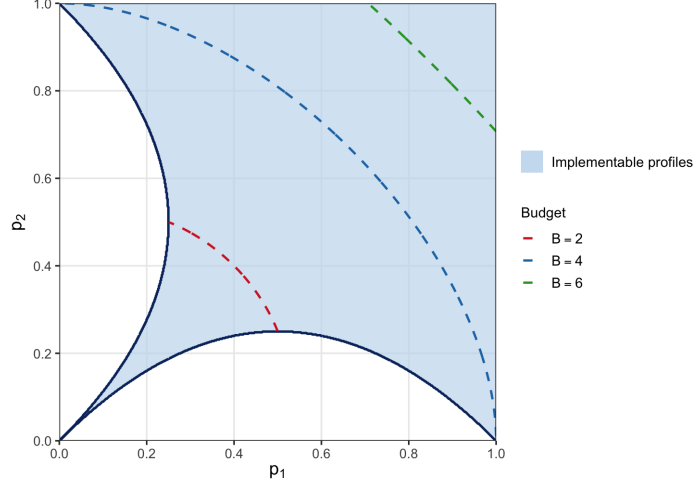


Figure 1: Effort profiles implementable by Luce contracts for $n = 2$, $c_i(p_i) = 2p_i^2$.

appropriate parameter range, this profile is $p = (\frac{v_1}{2C_1}, \frac{v_2}{2C_2})$. Applying Proposition 3, this profile is implementable by a Luce contract if and only if $\frac{v_i}{2C_i} \geq 1 - \frac{v_{-i}}{v_i}$ for each $i \in \{1, 2\}$. In particular, the optimal profile is always implementable when $v_1 = v_2$, regardless of C_1 and C_2 . A specific instance in which it is not implementable is $C_1 = C_2 = 4$ and $v_1 = 4$, $v_2 = 1$, yielding the optimal profile $(0.5, 0.125)$ —visible as a point outside the shaded region in Figure 1.

When possible, Luce contracts provide a particularly desirable alternative for implementation over other contracts by mitigating the uncertainty in total payments. For a contract f with budget B and a profile p , denote by $T_{f,B} = \sum_i f_i(S) \cdot B$ the (random) total payment.

Theorem 3. Suppose $p \in (0, 1)^n$ is implemented by a Luce contract f with budget B and an FGN contract g with budget B' . Then $T_{g,B'}$ is a mean preserving spread of $T_{f,B}$ at p .

It follows that among all contracts implementing a given effort profile, a Luce contract minimizes the variance of the principal's total payment. We note, however, that while Luce contracts reduce uncertainty for the principal, they introduce additional uncertainty for the agents: unlike a piece-rate contract, where an agent's payment depends only on their own success, payment under a Luce contract depends on the full set of successful agents, which may make them less transparent from the agents' perspective.⁷

Beyond minimizing variance, Luce contracts also minimize the maximum total payment the principal might have to make. The following example illustrates how large this reduction can be relative to piece-rate contracts, which are the natural choice of contract with independent agents and no budget constraint.

⁷We thank an anonymous referee for pointing this out.

Example 2. Consider n agents with identical quadratic costs, $c_i(p_i) = \frac{1}{2}Cp_i^2$. Suppose the principal's desired effort profile is such that all agents have the same success probability, $p_i = q$. This profile is implemented by the piece-rate contract which gives every agent Cq if they succeed and nothing otherwise, and it is also implemented by the Luce contract which splits a budget of $B = \frac{nCq^2}{1-(1-q)^n}$ equally among all successful agents. Both contracts yield the same expected total payment of nCq^2 , but the maximum possible total payment is nCq under piece-rate and $\frac{nCq^2}{1-(1-q)^n}$ under the Luce contract. In particular, if the principal wants there to be one successful agent in expectation, as might be the case for crowdsourcing an innovation, it would implement $q = \frac{1}{n}$, and the total payment could be as large as C under the piece-rate contract, where as the maximum possible total payment approaches 0 for large n under the Luce contract.

5 Conclusion

We study a contract design problem with multiple independent agents. With binary outcomes for each agent and a budget-constrained principal, we introduce a novel class of contracts, called Luce contracts, and demonstrate their sufficiency for implementing the maximal set of implementable equilibria. It follows that for any monotone objective of the principal, there is always a Luce contract that is optimal. Additionally, we highlight the relevance of Luce contracts in the absence of budget constraints, as they provide a means for inducing effort while reducing the uncertainty in total payments and in particular, mitigating the risk of excessively large payouts.

Our results and analysis open several promising directions for future research. Within the framework of our model with binary outcomes, an intriguing question is how the structure of the optimal Luce contract varies with heterogeneity in agents' costs under different objectives of the principal. This is particularly compelling in light of the comparative static observed in the special case of two agents with quadratic costs, where the principal strengthens incentives for the more preferred agent as the environment becomes more difficult. Another valuable avenue would be to extend our model to allow for multiple outcomes per agent and to investigate potential generalizations of Luce contracts in these richer and more complex environments.

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A Proofs for Section 3 (Main results)

Proposition 1. *For any implementable profile p , there exists an FGN contract g which implements it. Moreover, the expected total payment at p under g is minimal:*

$$\mathbb{E}_p\left[\sum_{i \in [n]} g_i(S)\right] \cdot B = \sum_{i \in [n]} p_i c'_i(p_i).$$

Proof. Let $p \in \mathcal{E}$ and let $f \in E^{-1}(p)$. Consider the contract g defined by

$$g_i(S) = \begin{cases} \lambda_i f_i(S), & \text{if } i \in S \\ 0, & \text{otherwise} \end{cases}$$

where

$$\lambda_i = \begin{cases} \frac{\mathbb{E}_p[f_i(S) \mid i \in S] - \mathbb{E}_p[f_i(S) \mid i \notin S]}{\mathbb{E}_p[f_i(S) \mid i \in S]} & \text{if } p_i > 0 \\ 0 & \text{if } p_i = 0. \end{cases}$$

It follows that $\lambda_i \leq 1$ so that g is indeed an FGN contract. Now if $p_i > 0$, agent i 's marginal utility at profile p under the contract g is

$$\begin{aligned} \frac{\partial u_i(p)}{\partial p_i} &= \mathbb{E}_p[g_i(S) \mid i \in S] \cdot B - c'_i(p_i) && \text{(Equation (1))} \\ &= \lambda_i \mathbb{E}_p[f_i(S) \mid i \in S] \cdot B - c'_i(p_i) \\ &= \mathbb{E}_p[f_i(S) \mid i \in S] \cdot B - \mathbb{E}_p[f_i(S) \mid i \notin S] \cdot B - c'_i(p_i) \\ &= 0. && \text{(because } p \in E(f)) \end{aligned}$$

By concavity of payoffs, $b_i(g, p_{-i}) = p_i$. If $p_i = 0$, then $g_i(S) = 0$ for all S , so $b_i(g, p_{-i}) = 0 = p_i$. Hence, $p \in E(g)$. It is straightforward to verify that the expected total payment under g at p is $\sum_{i=1}^n p_i c'_i(p_i)$, and it is weakly lower than any other f that may induce p . $\square$

Theorem 1. *Suppose the contract f implements the effort profile p . Then, p is maximal if and only if f is an SGE contract.*

Proof. Suppose f is a SGE contract and $p \in E(f)$. We want to show that p is maximal. Observe that

$$\begin{aligned} \frac{\partial z}{\partial x_i} \Big|_{x \geq p} &= B \prod_{j \neq i} (1 - x_j) - x_i c''_i(x_i) - c'_i(x_i) \\ &< B \prod_{j \neq i} (1 - x_j) - c'_i(x_i) && \text{(because } c_i \text{ is convex)} \\ &\leq B \prod_{j \neq i} (1 - x_j) - c'_i(p_i) && \text{(because } c_i \text{ is convex)} \end{aligned}$$

$$\begin{aligned}
&\leq B \prod_{j \neq i} (1 - p_j) - c'_i(p_i) && \text{(because } x_j \geq p_j) \\
&\leq B \cdot \mathbb{E}_p[f_i(S) | i \in S] - c'_i(p_i) && \text{(because } f \in \mathcal{F}_{SGE}) \\
&= 0 && \text{(because } p \in E(f))
\end{aligned}$$

It follows that if effort profile q dominates p , then $z(q) < z(p)$, and from Corollary 1, q is not implementable. Thus, p is maximal.

Now suppose p is maximal. We want to show that any $f \in E^{-1}(p)$ must be SGE. We first show that all agents must be active at p .

Lemma 2. *For all $i \in [n]$, $0 < p_i < 1$.*

Proof. Suppose towards a contradiction that $p_k = 0$ for some $k \in [n]$. Let $f \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ and consider the contract g where $g_i(S) = f_i(S \setminus \{k\})$ for $i \neq k$ and

$$g_k(S) = \begin{cases} 1, & \text{if } S = \{k\} \\ 0, & \text{otherwise} \end{cases}$$

It is easy to verify that $p' = (p'_k, p_{-k})$ where $p'_k = b_k(g, p_{-k}) > 0$ is an equilibrium under g . But then, $p' \in \mathcal{E}$ dominates p , which contradicts the fact that p is maximal. $\square$

Now let

$$K_p := \{S \subseteq [n] : \sum_{i \in S} f_i(S) < 1 \text{ for some } f \in E^{-1}(p) \cap \mathcal{F}_{FGN}\},$$

and let

$$\kappa_p := \{i \in [n] : \{i\} \in K_p\}.$$

We now show that $K_p = \{\emptyset\}$.

1. Step 1: Suppose $S \in K_p$. For any $T \subset S$, $T \in K_p$.

Let $f \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ be such that $\sum_{i \in S} f_i(S) < 1$. If $\sum_{i \in T} f_i(T) < 1$, we are done. Otherwise, pick an agent $i \in T$ such that $f_i(T) > 0$ and consider a contract g which differs from f only in its award for agent i at S and T . In particular, let g be such that $g_i(S) = f_i(S) + \epsilon$ and $g_i(T) = f_i(T) - \delta$ where $\epsilon, \delta > 0$ are chosen so that $p \in E(g)$. We can do this because we know from Lemma 2 that all events occur with positive probability at profile p . It follows then that $g \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ and $\sum_{i \in T} g_i(T) < 1$. Thus, $T \in K_p$.

2. Step 2: Suppose $S, T \in K_p$. Then, $S \cup T \in K_p$.

Let $f, g \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ be such that $\sum_{i \in S} f_i(S) < 1$ and $\sum_{i \in T} g_i(T) < 1$. Consider the contract $h = \frac{1}{2}f + \frac{1}{2}g$. From Equation (3), $h \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ and also

$\sum_{i \in S} h_i(S) < 1$ and $\sum_{i \in T} h_i(T) < 1$. Now, if $\sum_{i \in S \cup T} h_i(S \cup T) < 1$, we are done. Otherwise, pick any agent $i \in S \cup T$ (WLOG, let $i \in S$) such that $h_i(S \cup T) > 0$ and consider a contract h' which differs from h only in its award for agent i at $S \cup T$ and S . In particular, let h' be such that $h'_i(S \cup T) = h_i(S \cup T) - \epsilon$ and $h'_i(S) = h_i(S) + \delta$ where $\epsilon, \delta > 0$ are chosen so that $p \in E(h')$. It follows then that $h' \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ and $\sum_{i \in S \cup T} h'_i(S \cup T) < 1$. Thus, $S \cup T \in K_p$.

Note that it follows from Steps 1 and 2 that $K_p = 2^{\kappa_p}$.

3. Step 3: Suppose $f \in E^{-1}(p) \cap \mathcal{F}_{FGN}$. Then, for all $S \subset [n]$ such that $\kappa_p^C \cap S \neq \emptyset$, $f_i(S) = 0$ for all $i \in \kappa_p$.

Suppose towards a contradiction that there is an $S \subset [n]$ such that $\kappa_p^C \cap S \neq \emptyset$ and $f_i(S) > 0$ for some $i \in \kappa_p$. Let $g \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ be such that $g_i(\{i\}) < 1$. Consider the contract $h = \frac{1}{2}f + \frac{1}{2}g$. Then $h \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ and also $h_i(S) > 0$ and $h_i(\{i\}) < 1$. Now, consider a contract h' which differs from h only in its award for agent i at S and $\{i\}$. In particular, let h' be such that $h'_i(\{i\}) = h_i(\{i\}) + \epsilon$ and $h'_i(S) = h_i(S) - \delta$ where $\epsilon, \delta > 0$ are chosen so that $p \in E(h')$. It follows then that $h' \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ and $\sum_{i \in S} h'_i(S) < 1$. But this means that $S \subset \kappa_p$ which is a contradiction.

4. Step 4: Suppose $\kappa_p \neq \emptyset$. Then there is a $p' \in \mathcal{E}$ that dominates p .

For all $S \in K_p$, let $f^S \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ be such that $\sum_{i \in S} f_i^S(S) < 1$. Consider the contract $g = \sum_{S \in K_p} \frac{1}{|K_p|} f^S$. Then $g \in E^{-1}(p) \cap \mathcal{F}_{FGN}$ and also $\sum_{i \in S} g_i(S) < 1$ for all $S \in K_p$. Using g , we now construct a new contract

$$h_i(T) = \begin{cases} g_i(T), & \text{if } T \in K_p \\ \mathbb{E}_p[g_i(S) \mid S \setminus \kappa_p = T \setminus \kappa_p], & \text{if } T \notin K_p \end{cases}$$

Observe that if $S \notin K_p$ then $h_i(S) = h_i(S \setminus \kappa_p)$ for every $S \subseteq [n]$. Hence, if we manipulate h at any $S \subset \kappa_p$, it won't change the best responses for agents $i \notin \kappa_p$. We will now show that we can manipulate the awards for $S \subset \kappa_p$ so that under the new contract h' , $p' \in E(h')$ where $p'_i > p_i$ for $i \in \kappa_p$ and $p'_i = p_i$ for $i \notin \kappa_p$. Towards this goal, let $A = \kappa_p$ and let $p' = (p_i + \epsilon)_{i \in A}$. For each $i \in A$, let $t_i(\epsilon)$ solve

$$c'_i(p'_i) = B \cdot t_i(\epsilon) \cdot \sum_{S \subset A - i} (h_i(S \cup \{i\})) \Pr_{p'_{-i}}^{A-i}(S)$$

Observe that as $\epsilon \rightarrow 0$, $t_i(\epsilon) \rightarrow 1$ for all $i \in A$. Since $\sum_{i \in S} h_i(S) < 1$ for all $S \subset A$ and $t_i(\epsilon)$ is continuous in ϵ , we can find $\epsilon > 0$ small enough so that the contract $h'_i(S) = h_i(S) * t_i(\epsilon)$ for all $S \subset A$ and $i \in S$ is a feasible contract. By construction, the profile (p', p_{-A}) will be an equilibrium under h' . Thus, we have that p is not maximal.

It follows then that $K_p = \{\emptyset\}$. By definition of K_p , this means that for any $f \in E^{-1}(p)$, either $f \notin \mathcal{F}_{FGN}$ or $f \in \mathcal{F}_{SGE}$. Now suppose there exists an f such that $f \notin \mathcal{F}_{FGN}$ and $p \in E(f)$. From Proposition 1, we can find $g \in \mathcal{F}_{FGN}$ such that $p \in E(g)$. Moreover, we

know from the construction of g in the argument of Proposition 1 that $g \notin \mathcal{F}_{SGE}$. Thus, we have that $g \in \mathcal{F}_{FGN}$, $g \notin \mathcal{F}_{SGE}$, and $p \in E(g)$. But this means that $K_p \neq \emptyset$ which is a contradiction. Therefore, it must be that for any $f \in E^{-1}(p)$, $f \in \mathcal{F}_{SGE}$.

□

Theorem 2. *For every maximal equilibrium p , there is a unique contract in the closure of weighted contracts which implements p .*

Proof. From Theorem 1, we know that p must be an equilibrium of a SGE contract. But then, it follows that p must satisfy (4). From Proposition 3, there is a Luce contract that implements p . Thus, we get that every maximal equilibrium can be implemented by a Luce contract.

Now we show that there is a unique Luce contract that implements p . Consider the case where f, g are weighted contracts and suppose towards a contradiction that $p \in E(f) \cap E(g)$. Let $i \in [n]$ denote the agent with the smallest weight ratio $\frac{\lambda_i^g}{\lambda_i^f}$, where λ^f, λ^g represent the weights that define the contracts $f, g \in \mathcal{F}_W$. It follows from the definition of weighted contracts that for any $S \subset [n]$ such that $i \in S$, agent i 's reward is weakly lower under g than f . As a result, it can't be that $b_i(g, p_{-i}) = b_i(f, p_{-i})$. It follows that two different weighted contracts cannot have the same equilibrium.

Now suppose f, g are arbitrary Luce contracts that implement p . If $\succsim_f = \succsim_g$, then the same argument works. So suppose that $\succsim_f \neq \succsim_g$. Then there is some i such that $T = \{k : k \succsim_f i\} \neq \{k : k \succsim_g i\} = T'$, and we may assume without loss of generality that $T' \not\subseteq T$ by exchanging f and g if necessary. Observe that for any S ,

$$\sum_{k \in T} g_k(S) \leq \mathbb{1}_{S \cap T' \neq \emptyset} = \sum_{k \in T} f_k(S).$$

Moreover, for any $j \in T' \setminus T$,

$$\sum_{k \in T} g_k(\{i, j\}) = g_i(\{i, j\}) < 1 = f_i(\{i, j\}) = \sum_{k \in T} f_k(\{i, j\}).$$

Thus,

$$\sum_{k \in T} \mathbb{P}[k \in S] \cdot \mathbb{E}[g_k(S) \mid k \in S] = \mathbb{E}[\sum_{k \in T} g_k(S)] < \mathbb{E}[\sum_{k \in T} f_k(S)] = \sum_{k \in T} \mathbb{P}[k \in S] \cdot \mathbb{E}[f_k(S) \mid k \in S].$$

However, since p is an equilibrium for f and g , both of the outside expressions must be equal to $\frac{\sum_{k \in T} p_k \cdot c'_k(p_k)}{B}$, contradiction. □

Lemma 1. *A contract f is in the closure of weighted contracts if and only if it is a Luce contract.*

Proof. To begin, suppose that f is the limit of a sequence g^m of weighted contracts with weights $(\lambda_1^m, \dots, \lambda_n^m)$. Define the relation $\succeq$ by $i \succeq j$ if and only if $f_i(\{i, j\}) > 0$. It is immediate that $\succeq$ is complete. Moreover, since $f_i(\{i, j\}) > 0$ if and only if there exists an $\varepsilon > 0$ such that $\lambda_i^m \geq \varepsilon \lambda_j^m$ for all m , $\succeq$ is also transitive, and hence it is a weak order. For each i , let $S^i = \{j : i \sim j\}$, and define $\lambda_i = f_i(S^i)$. We claim that f is a Luce contract with priority ordering given by $\succeq$ and weights given by (λ_i) . To prove this, fix i and S .

If $i \notin S$, then

$$f_i(S) = \lim_{m \rightarrow \infty} g_i^m(S) = \lim_{m \rightarrow \infty} 0 = 0.$$

If $i \in S \setminus \text{Top}_{\succeq}(S)$, then there is some $j \in S$ such that $j \succ i$. Now,

$$g_i^m(S) = \frac{\lambda_i^m}{\sum_{k \in S} \lambda_k^m} \leq \frac{\lambda_i^m}{\lambda_i^m + \lambda_j^m} = g_i^m(\{i, j\}),$$

so

$$f_i(S) = \lim_{m \rightarrow \infty} g_i^m(S) \leq \lim_{m \rightarrow \infty} g_i^m(\{i, j\}) = f_i(\{i, j\}) = 0.$$

Finally, observe that for any i and j , if $i \succ j$, then

$$\lim_{m \rightarrow \infty} \frac{1}{1 + \frac{\lambda_j^m}{\lambda_i^m}} = \lim_{m \rightarrow \infty} \frac{\lambda_i^m}{\lambda_i^m + \lambda_j^m} = \lim_{m \rightarrow \infty} g_i^m(\{i, j\}) = f_i(\{i, j\}) = 1,$$

so $\lim_{m \rightarrow \infty} \frac{\lambda_j^m}{\lambda_i^m} = 0$, and if $i \sim j$, then $S^i = S^j$, so

$$\lim_{m \rightarrow \infty} \frac{\lambda_j^m}{\lambda_i^m} = \lim_{m \rightarrow \infty} \frac{\frac{\lambda_j^m}{\sum_{k \in S^j} \lambda_k^m}}{\frac{\lambda_i^m}{\sum_{k \in S^i} \lambda_k^m}} = \lim_{m \rightarrow \infty} \frac{g_j^m(S^j)}{g_i^m(S^i)} = \frac{f_j(S^j)}{f_i(S^i)} = \frac{\lambda_j}{\lambda_i}.$$

Hence, if $i \in \text{Top}_{\succeq}(S)$, then

$$\begin{aligned} f_i(S) &= \lim_{m \rightarrow \infty} g_i^m(S) = \lim_{m \rightarrow \infty} \frac{\lambda_i^m}{\sum_{k \in S} \lambda_k^m} = \lim_{m \rightarrow \infty} \frac{1}{\sum_{k \in \text{Top}_{\succeq}(S)} \frac{\lambda_k^m}{\lambda_i^m} + \sum_{k \in S \setminus \text{Top}_{\succeq}(S)} \frac{\lambda_k^m}{\lambda_i^m}} \\ &= \frac{1}{\sum_{k \in \text{Top}_{\succeq}(S)} \frac{\lambda_k}{\lambda_i}} = \frac{\lambda_i}{\sum_{k \in \text{Top}_{\succeq}(S)} \lambda_k}. \end{aligned}$$

Since f was arbitrary, it follows that $\bar{F}_W \subseteq \mathcal{F}_{\text{Luce}}$.

Now, suppose f is a Luce contract. For each i , let $r_i = |\{j : j \succeq i\}|$, and for every $t > 0$, let g^t be the weighted contract with weights given by $\lambda_i^t = t^{r_i} \cdot \lambda_i$. Now, fix i and S . If $i \notin S$,

$$\lim_{t \rightarrow 0} g_i^t(S) = \lim_{t \rightarrow 0} 0 = 0 = f_i(S).$$

If $i \in S \setminus Top_{\succeq}(S)$, then there is some $j \in S$ such that $j \succ i$ and hence $r_j < r_i$. In particular,

$$\lim_{t \rightarrow 0} g_i^t(S) = \lim_{t \rightarrow 0} \frac{t^{r_i} \cdot \lambda_i}{\sum_{k \in S} t^{r_k} \cdot \lambda_k} \leq \lim_{t \rightarrow 0} \frac{t^{r_i} \lambda_i}{t^{r_j} \lambda_j} = 0 = f_i(S).$$

Finally, if $i \in Top_{\succeq}(S)$, then

$$\begin{aligned} g_i^t(S) &= \frac{t^{r_i} \lambda_i}{\sum_{k \in S} t^{r_k} \cdot \lambda_k} = \frac{t^{r_i} \lambda_i}{t^{r_i} \cdot \sum_{k \in Top_{\succeq}(S)} \lambda_k + \sum_{k \in S \setminus Top_{\succeq}(S)} t^{r_k} \cdot \lambda_k} \\ &= \frac{\lambda_i}{\sum_{k \in Top_{\succeq}(S)} \lambda_k + \sum_{k \in S \setminus Top_{\succeq}(S)} t^{r_k - r_i} \cdot \lambda_k}. \end{aligned}$$

Since $r_k > r_i$ for all $k \in S \setminus Top_{\succeq}(S)$, it follows that

$$\lim_{t \rightarrow 0} g_i^t(S) = \frac{\lambda_i}{\sum_{k \in Top_{\succeq}(S)} \lambda_k} = f_i(S).$$

Thus, $f = \lim_{t \rightarrow 0} g^t$. Since f was arbitrary, it follows that $\mathcal{F}_{Luce} \subseteq \bar{\mathcal{F}}_W$. $\square$

Proposition 2. Suppose $n = 2$, $c_i(p_i) = \frac{1}{2}C_i p_i^2$ with $C_i > B$, and $V(p_1, p_2) = v_1 p_1 + v_2 p_2$ with $v = \frac{v_1}{v_2} \geq 1$. Then, the optimal contract, defined by λ_1 , takes the form

$$f_i(S) = \begin{cases} 0, & \text{if } i \notin S \\ 1, & \text{if } S = \{i\} \\ \lambda_i, & \text{if } S = \{1, 2\} \end{cases},$$

where

$$\lambda_1 = \begin{cases} \frac{1}{2}, & \text{if } v = 1 \\ \frac{-(vC_2 + C_1 - vB) + \sqrt{(v^2C_2 + C_1 - vB)(C_2 + C_1 - B)}}{B(v - 1)}, & \text{if } v \in (1, \frac{C_1C_2 + BC_1 - B^2}{C_1C_2 - BC_2}) \\ 1, & \text{if } v \geq \frac{C_1C_2 + BC_1 - B^2}{C_1C_2 - BC_2} \end{cases}$$

and $\lambda_2 = 1 - \lambda_1$. It follows that λ_1 is weakly increasing in v , C_1 , and C_2 , and weakly decreasing in B .

Proof. We know from Theorem 1 that we can restrict attention to SGE contracts. And with only two agents, the set of SGE contracts can be parametrized by a single parameter λ , where $f_i(S) = 0$ whenever $i \notin S$, $f_i(S) = 1$ whenever $S = \{i\}$, and $f_1(\{1, 2\}) = 1 - f_2(\{1, 2\}) = \lambda$. The equilibrium conditions for this contract from Equation (3) are

$$\begin{aligned} C_1 p_1 &= (1 - p_2 + \lambda p_2) \cdot B \\ C_2 p_2 &= (1 - p_1 + (1 - \lambda) p_1) \cdot B. \end{aligned}$$

Going forward, we normalize $C_i = \frac{C_i}{B}$. For each λ , this system of equations has a unique solution:

$$p_1(\lambda) = \frac{C_2 - (1 - \lambda)}{C_1 C_2 - \lambda(1 - \lambda)} \quad p_2(\lambda) = \frac{C_1 - \lambda}{C_1 C_2 - \lambda(1 - \lambda)}. \quad (5)$$

Hence, the principal's problem is equivalent to:

$$\max_{\lambda \in [0,1]} v_1 p_1(\lambda) + v_2 p_2(\lambda).$$

Using the first order condition, this is maximized either at $\lambda = 0$, or $\lambda = 1$, or where

$$\frac{p'_2(\lambda)}{p'_1(\lambda)} = -\frac{v_1}{v_2}.$$

Note that

$$p'_1(\lambda) = \frac{(C_1 C_2 - \lambda(1 - \lambda)) - (C_2 - (1 - \lambda))(2\lambda - 1)}{(C_1 C_2 - \lambda(1 - \lambda))^2} = \frac{C_1 C_2 - C_2(2\lambda - 1) - (1 - \lambda)^2}{(C_1 C_2 - \lambda(1 - \lambda))^2}$$

and

$$p'_2(\lambda) = \frac{-(C_1 C_2 - \lambda(1 - \lambda)) - (C_1 - \lambda)(2\lambda - 1)}{(C_1 C_2 - \lambda(1 - \lambda))^2} = \frac{-C_1 C_2 - C_1(2\lambda - 1) + \lambda^2}{(C_1 C_2 - \lambda(1 - \lambda))^2}$$

so

$$\frac{p'_2(\lambda)}{p'_1(\lambda)} = \frac{-C_1 C_2 - C_1(2\lambda - 1) + \lambda^2}{C_1 C_2 - C_2(2\lambda - 1) - (1 - \lambda)^2} = -\frac{C_1}{C_2} \cdot \frac{-\lambda^2/C_1 + 2\lambda + C_2 - 1}{-(1 - \lambda)^2/C_2 + 2(1 - \lambda) + C_1 - 1}$$

Putting together, we get that λ must be a solution to the quadratic equation

$$y(\lambda) = \lambda^2(v_1 - v_2) - 2\lambda(v_1 - v_1 C_2 - v_2 C_1) + v_1 - v_1 C_2 - v_2 C_1 - C_1 C_2(v_1 - v_2) = 0.$$

Notice that since $v_1 \geq v_2$, $y(0) < 0$. Thus, λ must be equal to the larger root. Further, observe that

$$y(1) \geq 0 \iff \frac{v_1}{v_2} \leq \frac{C_1 C_2 + C_1 - 1}{C_1 C_2 - C_2}.$$

When $\frac{v_1}{v_2} = 1$, it is easy to see that $\lambda = \frac{1}{2}$ is the unique solution, irrespective of C_1, C_2 .

When $\frac{v_1}{v_2} \geq \frac{C_1 C_2 + C_1 - 1}{C_1 C_2 - C_2}$, $y(1) \leq 0$ so that the larger root is ≥ 1 , implying $\lambda = 1$.

When $\frac{v_1}{v_2} \in (1, \frac{C_1 C_2 + C_1 - 1}{C_1 C_2 - C_2})$, we can solve the quadratic to get that the larger root takes the form in the result. □

B Proofs for Section 4 (Luce contracts without budget constraints)

Proposition 3. *Suppose $p \in (0, 1)^n$. There exists a Luce contract $f \in \mathcal{F}_{Luce}$ with budget $B > 0$ that implements p if and only if for all $I \subset [n]$,*

$$\sum_{i \in I} p_i \cdot c'_i(p_i) \leq \mathbb{P}_p[S \cap I \neq \emptyset] \cdot B, \quad (4)$$

where $B = \frac{\sum_{i \in [n]} p_i \cdot c'_i(p_i)}{\mathbb{P}_p[S \neq \emptyset]}$.

Proof. Suppose $f \in \mathcal{F}_{Luce}$ with budget $B > 0$ implements p . Then for any $I \subseteq [n]$, the expected total reward to agents in I is

$$\begin{aligned} \sum_{i \in I} p_i \cdot c'_i(p_i) &= \mathbb{E}_p \left[\sum_{i \in I} f_i(S) \right] \cdot B \\ &\leq \mathbb{P}_p[S \cap I \neq \emptyset] \cdot B \end{aligned}$$

with equality for $I = [n]$. Thus, p must satisfy (4).

Now, suppose $p \in (0, 1)^n$ satisfies (4). For any $f \in \mathcal{F}_{Luce}$ with budget B , let

$$Z(f) := \max_{i \in [n]} (c'_i(b_i(f, p_{-i})) - c'_i(p_i))$$

and

$$\mathcal{C}(f) := \{i \in [n] : c'_i(b_i(f, p_{-i})) - c'_i(p_i) = Z(f)\}.$$

From Equation (3),

$$\sum_{i \in [n]} p_i \cdot [c'_i(b_i(f, p_{-i})) - c'_i(p_i)] = 0.$$

Moreover, for any agent $i \in [n]$,

$$\sum_{j: j \succ i} p_j \cdot c'_j(b_j(f, p_{-j})) = B \cdot \mathbb{E}_p \left[\sum_{j: j \succ i} f_j(S) \right] = B \cdot \mathbb{P}_p[S \cap \{j : j \succ i\} \neq \emptyset] \geq \sum_{j: j \succ i} p_j \cdot c'_j(p_j),$$

where the last inequality follows by taking $I = \{j : j \succ i\}$ in (4). Hence,

$$\sum_{j: j \succ i} p_j \cdot [c'_j(b_j(f, p_{-j})) - c'_j(p_j)] \geq 0,$$

and thus

$$\sum_{j: i \succ j} p_j \cdot [c'_j(b_j(f, p_{-j})) - c'_j(p_j)] \leq 0.$$

It follows that $Z(f) \geq 0$ for all $f \in \mathcal{F}_{Luce}$ and so

$$z = \inf_{f \in \mathcal{F}_{Luce}} Z(f) \geq 0.$$

We will now show that $z = 0$.

Suppose towards a contradiction that $z > 0$. Let $f \in \mathcal{F}_{Luce}$ be such that

1. $Z(f) = z$ and
2. for any other $g \in \mathcal{F}_{Luce}$ such that $Z(g) = z$, $\mathcal{C}(g) \not\subset \mathcal{C}(f)$.

Let $(X_1, \dots, X_\ell)$ be the ordered partition corresponding to $\succsim$ and $\lambda_1, \dots, \lambda_n$ be the weights that define f .

Let k be the maximum index such that $X_k \cap \mathcal{C}(f) \neq \emptyset$.

First suppose $k = \ell$. From above, we know that there must be some $i \in X_\ell$ such that $c'_i(b_i(f, p_{-i})) - c'_i(p_i) < 0$. Now consider another Luce contract g which is the same as f , except that the weight for agent i is $\lambda'_i = \lambda_i + \epsilon$. Thus, when all agents in $[n] \setminus X_\ell$ have failed, agent i gets a slightly higher share of the reward if it succeeds under g than it did under f . Notice that for $j \notin X_\ell$,

$$c'_j(b_j(g, p_{-j})) - c'_j(p_j) = c'_j(b_j(f, p_{-j})) - c'_j(p_j)$$

and for $j \in X_\ell \setminus \{i\}$,

$$c'_j(b_j(g, p_{-j})) - c'_j(p_j) < c'_j(b_j(f, p_{-j})) - c'_j(p_j) \leq z.$$

Moreover, for ϵ sufficiently small,

$$c'_i(b_i(g, p_{-i})) - c'_i(p_i) < 0$$

by continuity. It follows that $Z(g) \leq z$. In particular, we either have $Z(g) < z$ or $Z(g) = z$ and $\mathcal{C}(g) \subset \mathcal{C}(f)$. In either case, we have a contradiction.

Thus, it must be that $k < \ell$. Now, consider another Luce contract g which has the partition $(X_1, \dots, X_{k-1}, X_k \cup X_{k+1}, X_{k+2}, \dots, X_\ell)$ and weights

$$\lambda'_i = \begin{cases} \lambda_i & \text{if } i \notin X_{k+1} \\ \epsilon \lambda_i & \text{if } i \in X_{k+1} \end{cases}.$$

Then for $j \notin X_k \cup X_{k+1}$,

$$c'_j(b_j(g, p_{-j})) - c'_j(p_j) = c'_j(b_j(f, p_{-j})) - c'_j(p_j)$$

and for $j \in X_k$,

$$c'_j(b_j(g, p_{-j})) - c'_j(p_j) < c'_j(b_j(f, p_{-j})) - c'_j(p_j) \leq z.$$

Finally, for ε sufficiently small and $j \in X_{k+1}$,

$$c'_j(b_j(g, p_{-j})) - c'_j(p_j) < z$$

by continuity. It follows that $Z(g) \leq z$. In particular, we either have $Z(g) < z$ or $Z(g) = z$ and $\mathcal{C}(g) \subset \mathcal{C}(f)$. In either case, we have a contradiction. Thus, we have that $\inf_{f \in \mathcal{F}_{Luce}} Z(f) = 0$. By compactness of $\mathcal{F}_{Luce}$, there exists an $f \in \mathcal{F}_{Luce}$ such that $Z(f) = 0$ which means

$$\left[\max_{i \in [n]} (c'_i(b_i(f, p_{-i})) - c'_i(p_i)) \right] = 0.$$

But from above, we also know that

$$\sum_{i \in [n]} p_i [c'_i(b_i(f, p_{-i})) - c'_i(p_i)] = 0.$$

Therefore, it must be that for all $i \in [n]$,

$$(c'_i(b_i(f, p_{-i})) - c'_i(p_i)) = 0$$

which implies $f \in E^{-1}(p)$. Thus, for any p that satisfies (4), there is a Luce contract f with budget B such that $p \in E(f)$. □

Theorem 3. Suppose $p \in (0, 1)^n$ is implemented by a Luce contract f with budget B and an FGN contract g with budget B' . Then $T_{g, B'}$ is a mean preserving spread of $T_{f, B}$ at p .

Proof. By Proposition 1, for any FGN contract h (with budget B) that implements p , $\mathbb{E}_p[T_{h, B}] = \sum_i p_i \cdot c'_i(p_i)$. Moreover, since h is FGN, $S = \emptyset \Rightarrow T_{h, B} = 0$ and $\mathbb{E}_p[T_{h, B}] = \mathbb{P}_p[S \neq \emptyset] \cdot \mathbb{E}_p[T_{h, B} | S \neq \emptyset]$, so $\mathbb{E}_p[T_{f, B} | S \neq \emptyset] = \mathbb{E}_p[T_{g, B'} | S \neq \emptyset]$.

Observe that $S \neq \emptyset \Rightarrow T_{f, B} = B$, and it follows that $\mathbb{E}_p[T_{g, B'} | S \neq \emptyset] = B$. Now, let $X = 0$ if $S = \emptyset$ and $X = T_{g, B'} - B$ otherwise. Then $T_{g, B'} = T_{f, B} + X$, $\mathbb{E}_p[X | T_{f, B} = 0] = \mathbb{E}_p[X | S = \emptyset] = 0$, and

$$\mathbb{E}_p[X | T_{f, B} = B] = \mathbb{E}_p[T_{g, B'} - B | S \neq \emptyset] = 0,$$

so $T_{g, B'}$ is a mean-preserving spread of $T_{f, B}$ at p . □